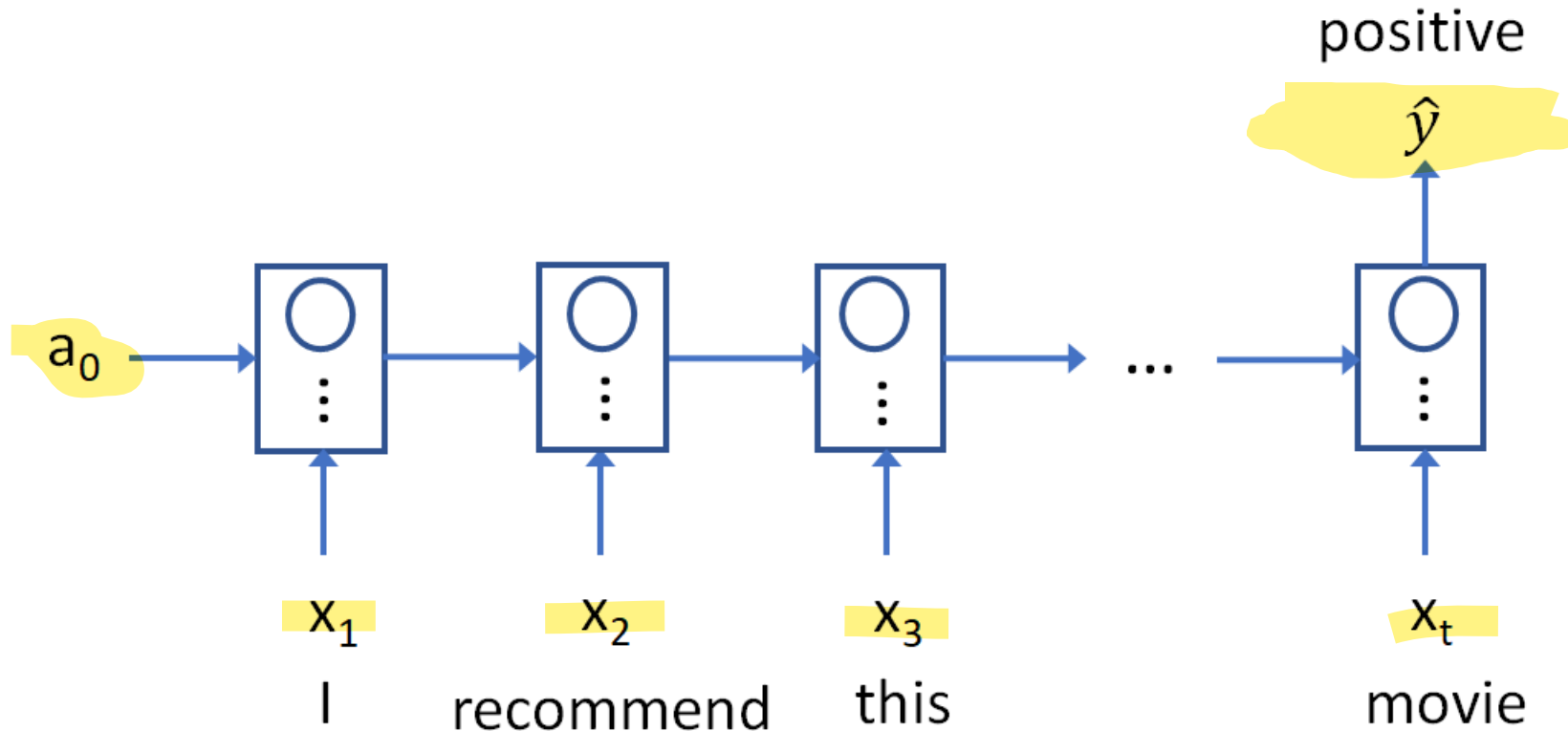


# Recurrent Neural Network

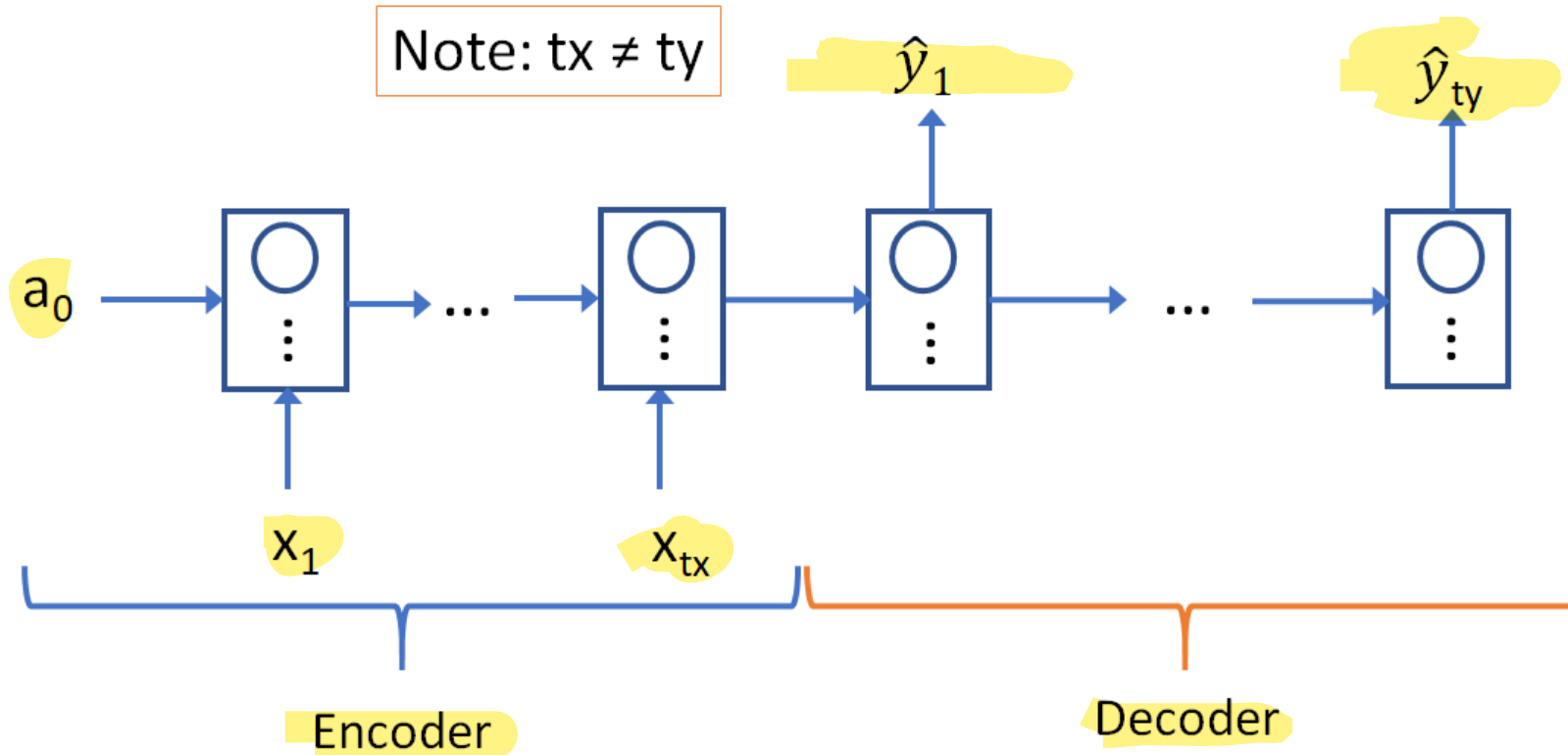
Selpi

# Many-to-one architecture



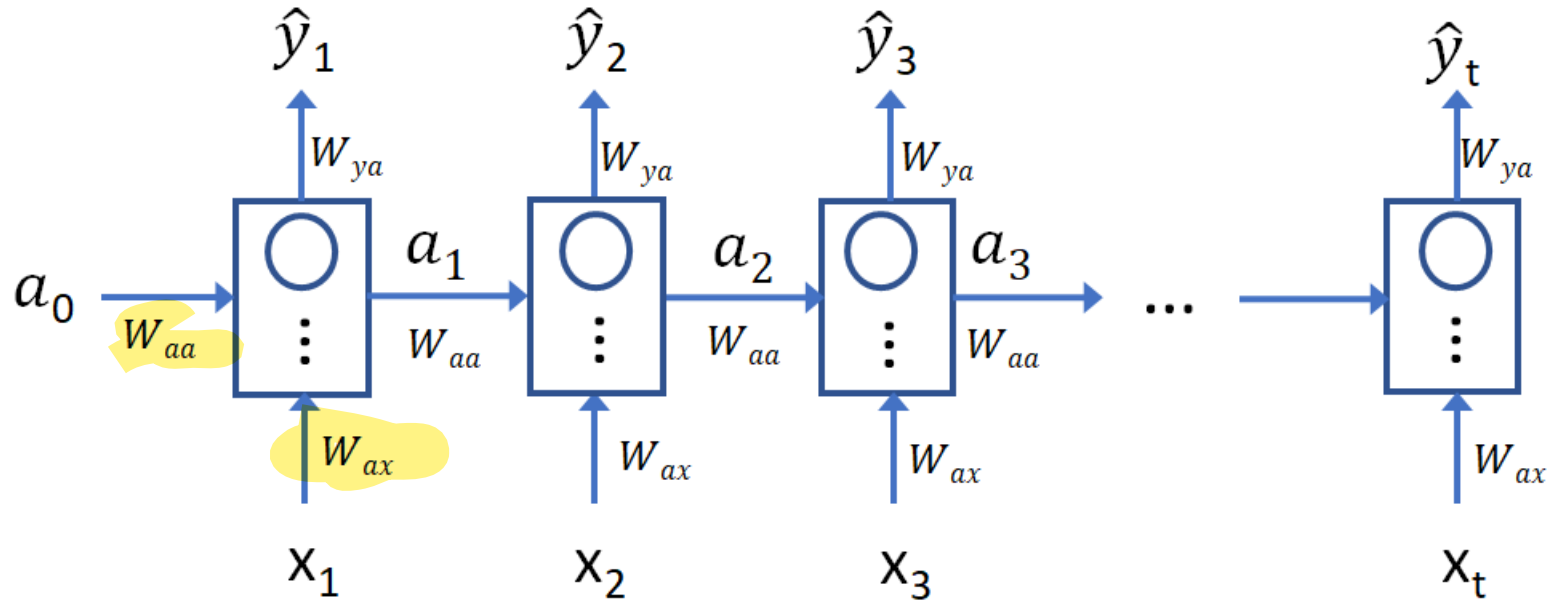
- Example: Sentiment classification, activity recognition

# Many-to-many architecture



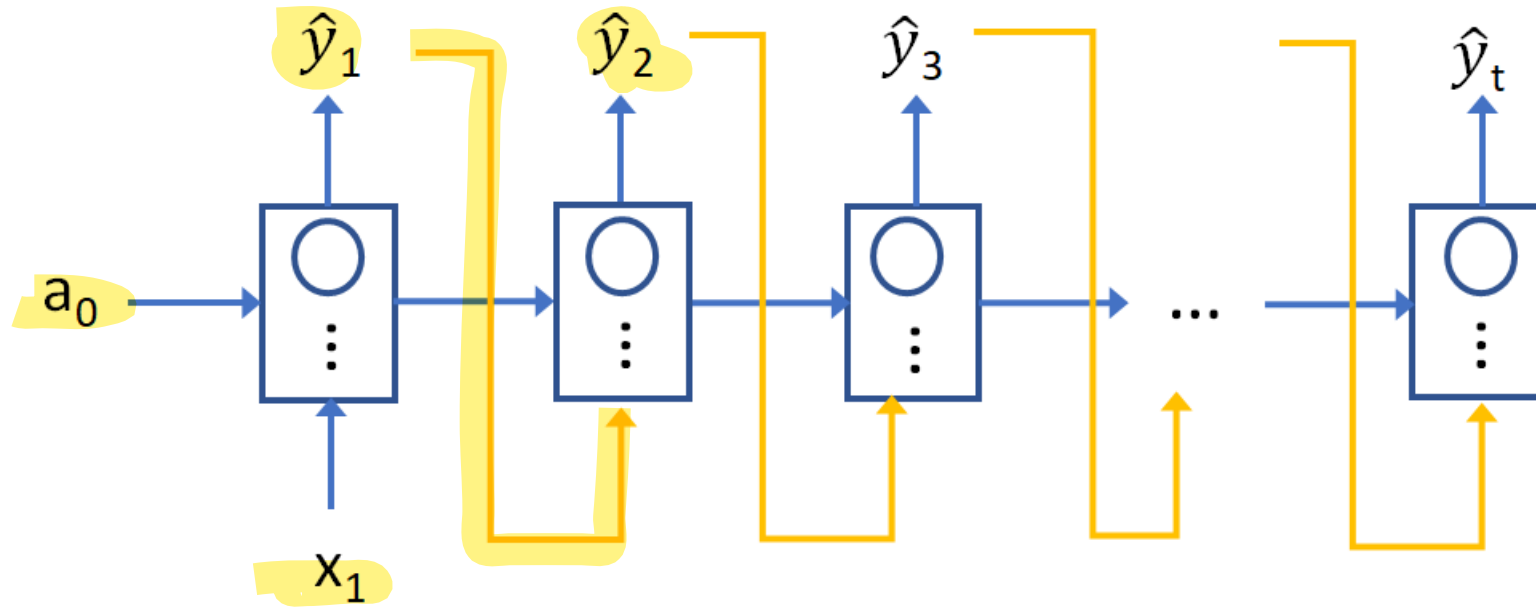
- For input and output sequences of the different length
- Example: machine translation

# Many-to-many architecture



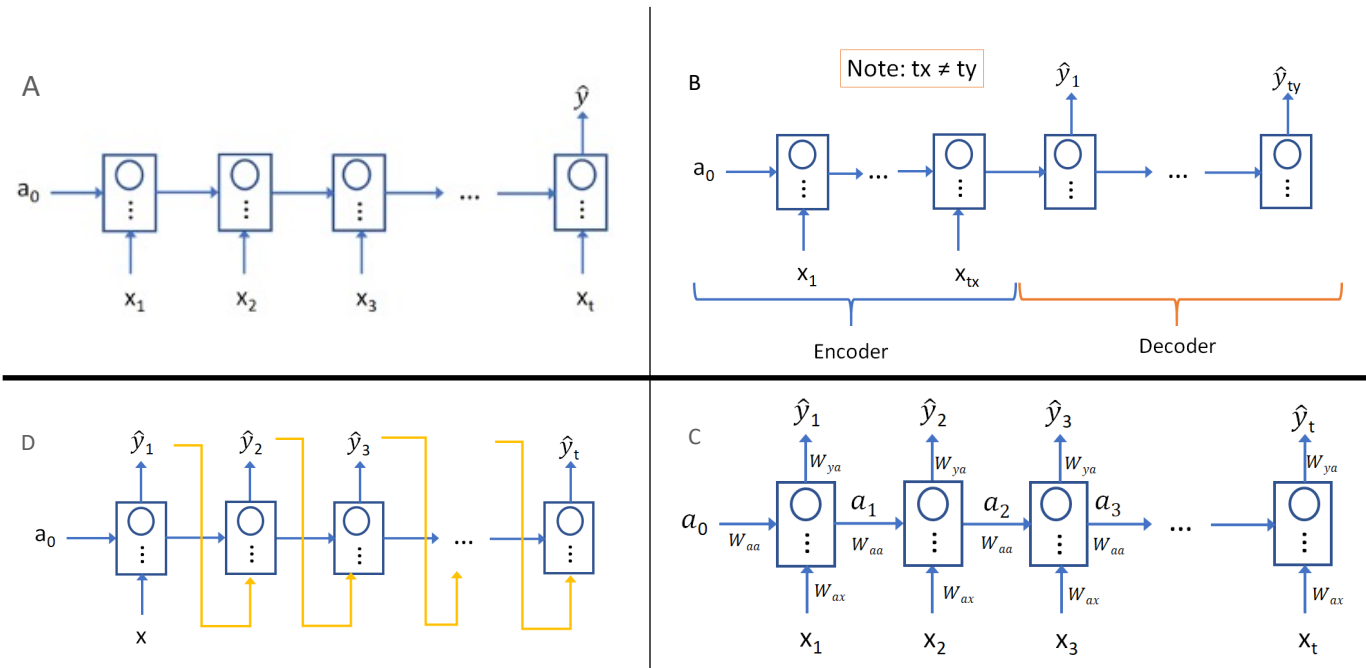
- For input and output sequences of the same length
- Example: video frame-by-frame classification, named entity recognition

# One-to-many architecture

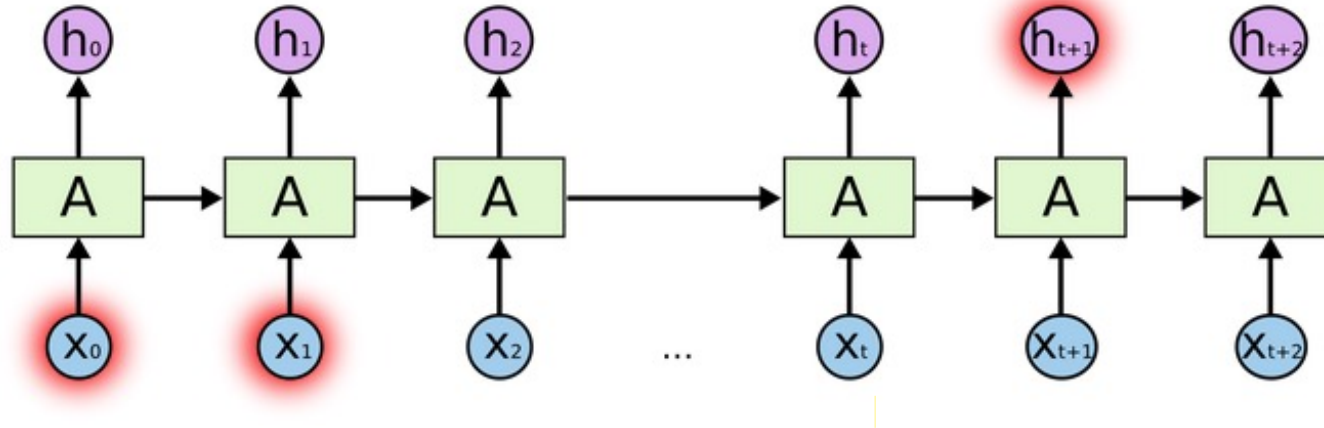


- Example: music generation, input: genre, output: sequence of notes

# Some RNN architectures



# The vanilla Recurrent Neural Network (RNN)



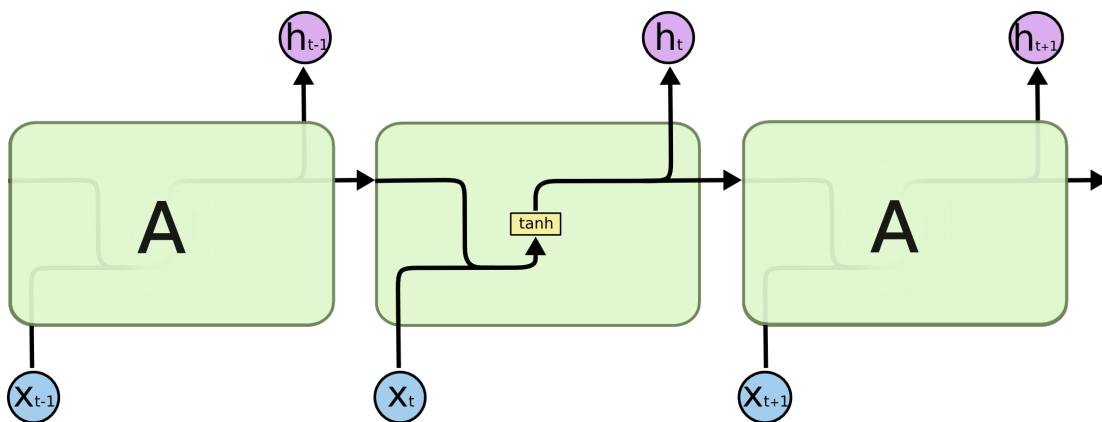
## Advantages:

- Can process input of any length
- Can take into account historical information
- Weights are shared across time
- Model size is not increasing with size of input

## Drawbacks:

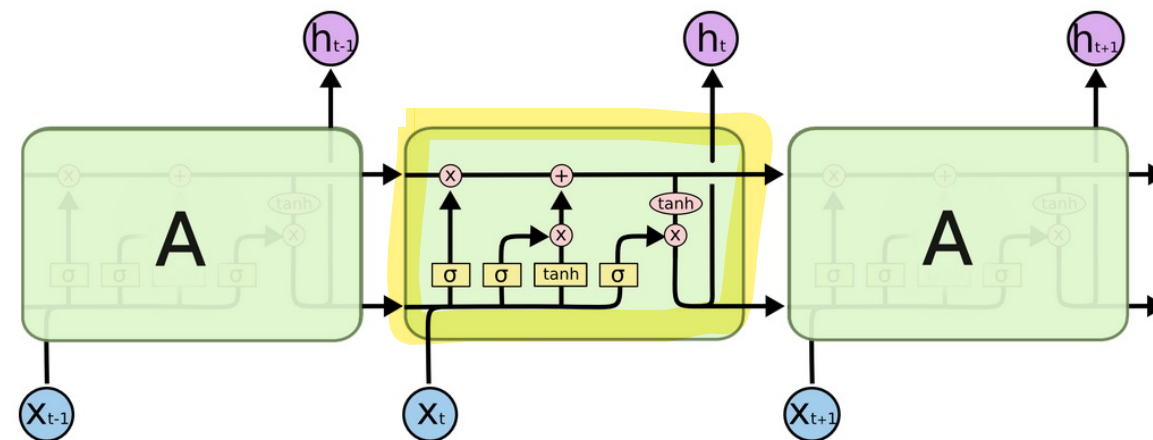
- Slow computation
- Information from a long time ago is not easy to access with vanilla RNN

# Vanilla RNN and LSTM



Vanilla RNN

- Can take into account historical information, but not good to keep info for long period of time

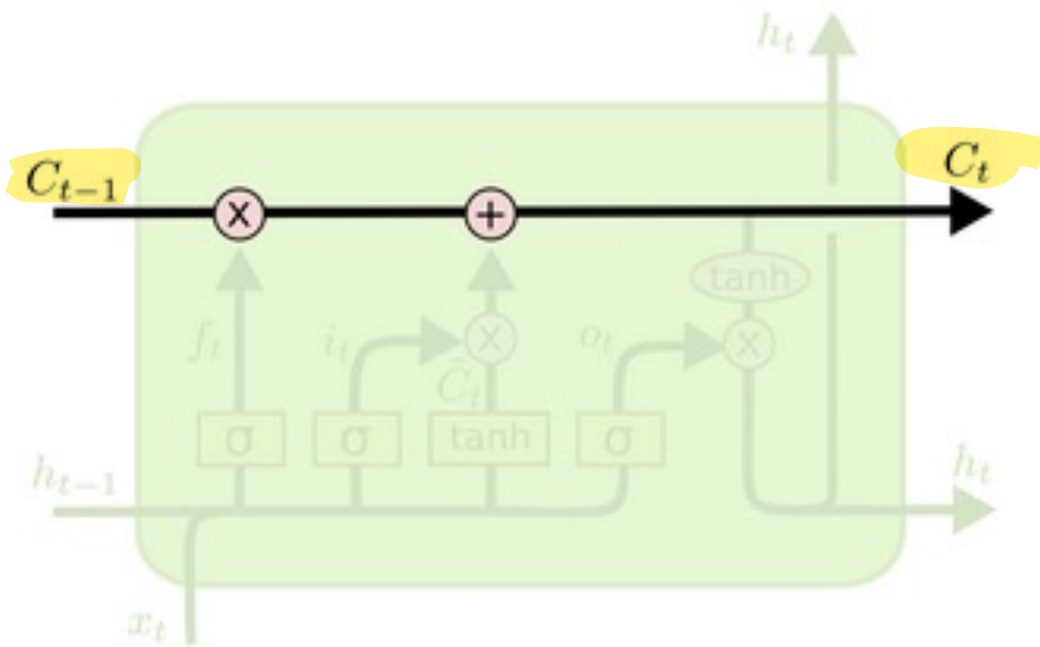


LSTM (Long short-term memory)

- Has memory cell thus can maintain info in memory for long period of time
- Has short term and long term parts

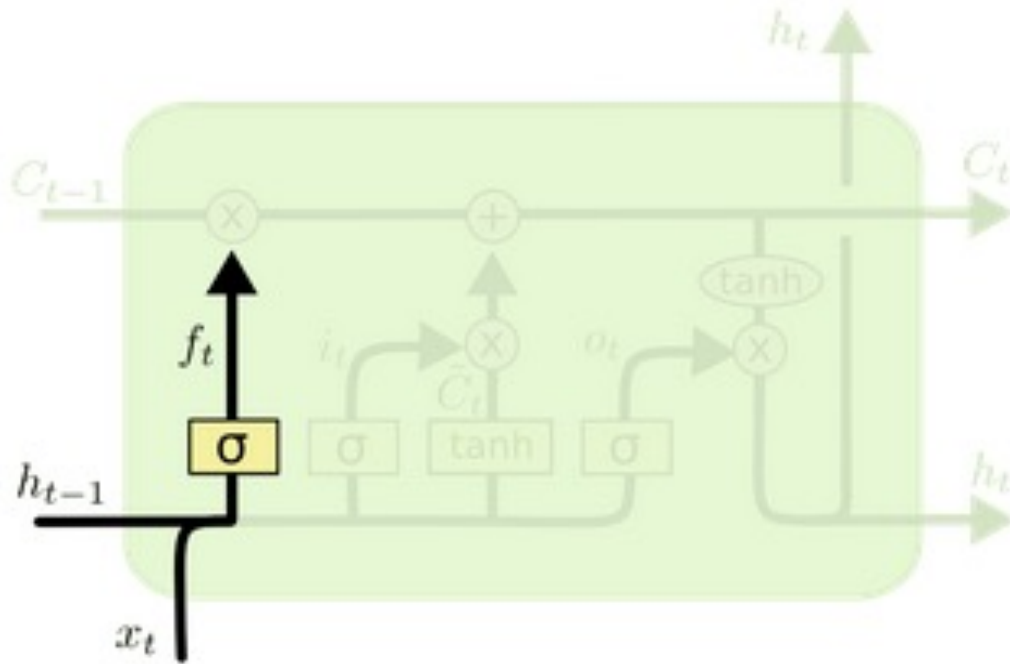


# LSTM



- Cell state, can easily pass through info unchanged

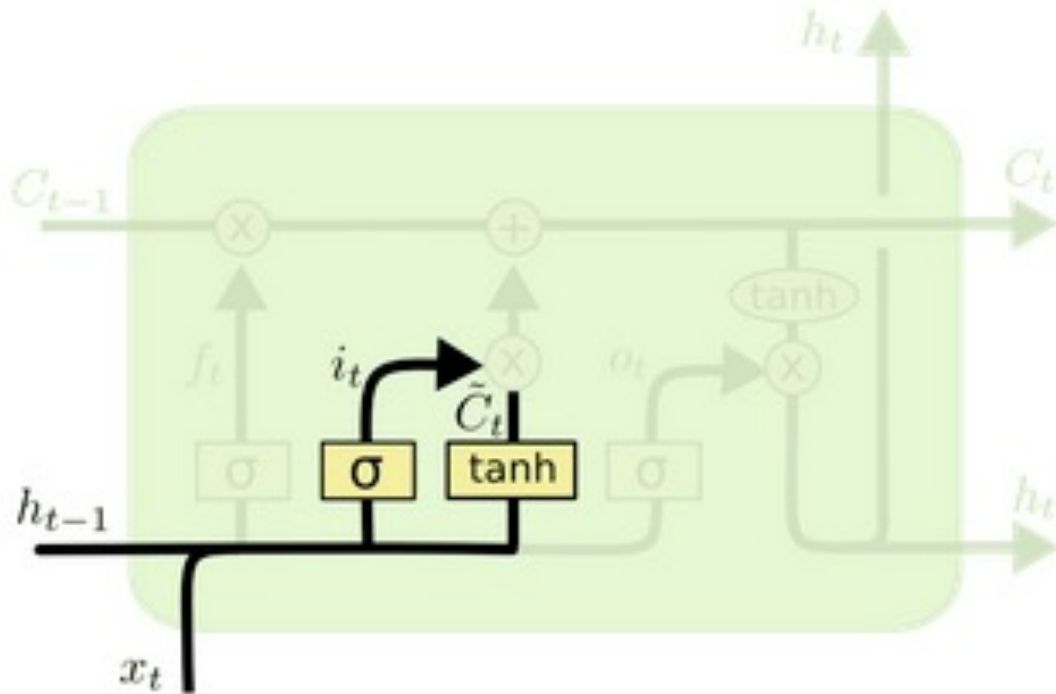
# LSTM



$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

- Forget gate decides what info should be kept/thrown away from the cell state.

# LSTM

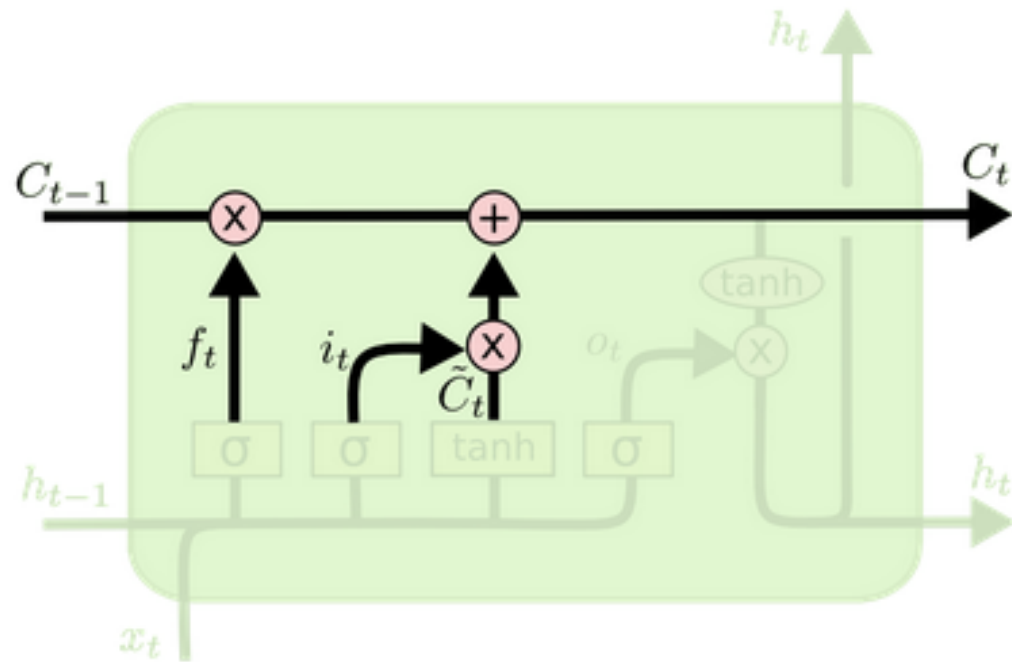


$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

- Input gate decides which values to update.

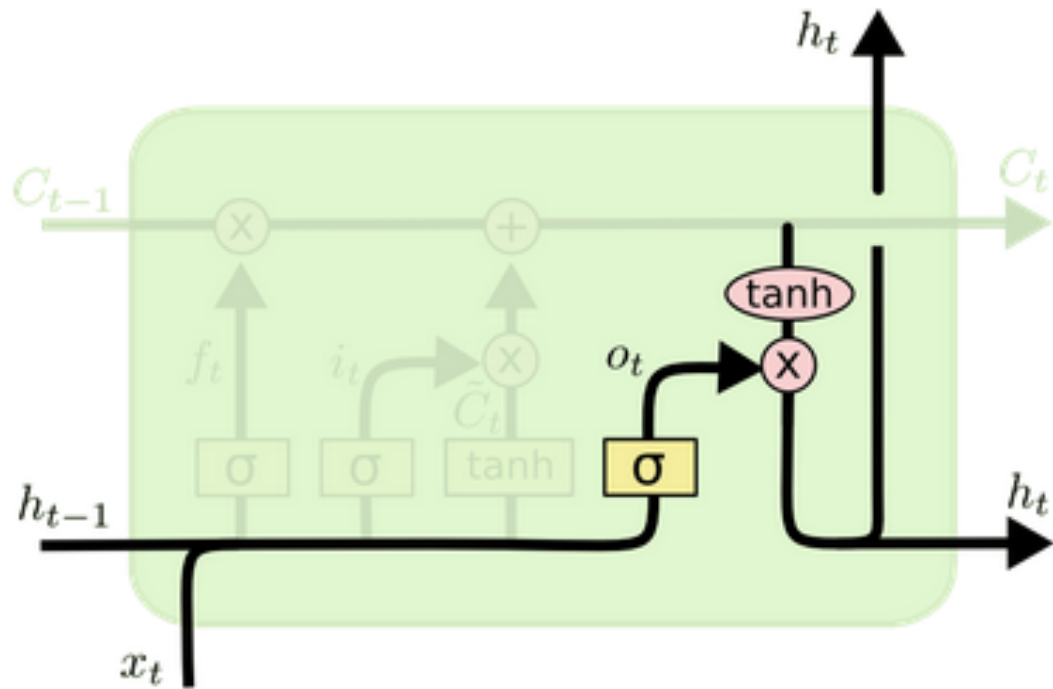
# LSTM



$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

- Update the old cell state,  $C_{t-1}$ , into the new cell state  $C_t$ .

# LSTM

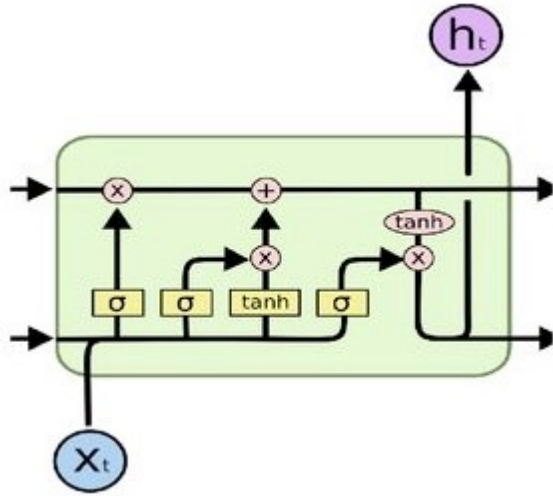


$$o_t = \sigma (W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh (C_t)$$

- The final output is based on a filtered version of the cell state and the decision from output gate in terms of which part to be sent to output.

# LSTM (left) and GRU (right)



$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

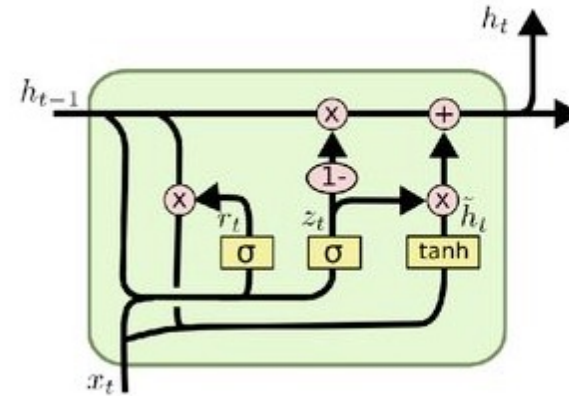
$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$



$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$